

MODULE 1

Introduction to Big Data

Module contents

Introduction to Big data, Conventional Data v/s Big data, Big data architecture, Big data platforms, Nature of data, Analytic processes and tools, 5 V's of Big data, Big data analytical method, Intelligent data analysis, Big data analytics life cycle.

Nature of Data

- Data is a set of value of qualitative and quantitative variables. In other words, pieces of data are individual pieces of information. Data is measured, collected and reported and analysed where upon it can be visualized using graph or images.

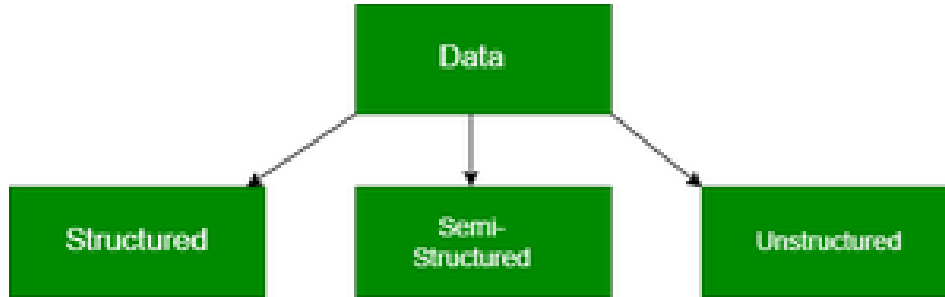
Properties of Data

- **Flexibility of use:** data are facts used in designing something. Data are meant to be used as a phase for arriving at definitive conclusion
- **Clarity:** data are crystallized presentation. without clarity, the meaning desired to be communicated will remain hidden

- **Accuracy:** data should be real, complete and accurate.
Accuracy is an essential property of data
- **Essence:** A large quantities of data are collected and they have to be compressed and refined. Data are refined can present the essence as derived qualitative value of the matter.
- **Aggregation:** It is meant by adding up data coming from different data sources

- **Compression:** Large amounts of data are always compressed to make them more meaningful. Compress data into a managed size. Graphs and charts are some examples of compressed data
- **Refinement:** data require processing or refinement. When refined, they are capable of leading conclusion or even generalization.

Types of Data

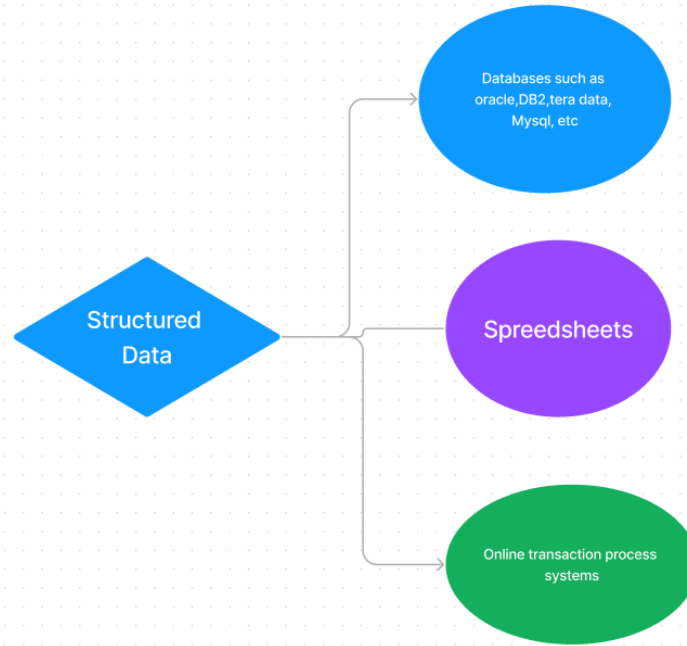


-
-

Structured Data

- Structured data can be crudely defined as the data that resides in a fixed field within a record.
- A certain schema binds it, so all the data has the same set of properties. Structured data is also called relational data. It is split into multiple tables to enhance the integrity of the data by creating a single record to depict an entity. Relationships are enforced by the application of table constraints.

- . The business value of structured data lies within how well an organization can utilize its existing systems and processes for analysis purposes.



Sources of structured data

A Structured Query Language (SQL) is needed to bring the data together. Structured data is easy to enter, query, and analyze. All of the data follows the same format. However, forcing a consistent structure also means that any alteration of data is too tough as each record has to be updated to adhere to the new structure. Examples of structured data include numbers, dates, strings, etc. The business data of an e-commerce website can be considered to be structured data.

Name	Class	Section	Roll No	Grade
Geek1	11	A	1	A
Geek2	11	A	2	B
Geek3	11	A	3	A

Cons of Structured Data

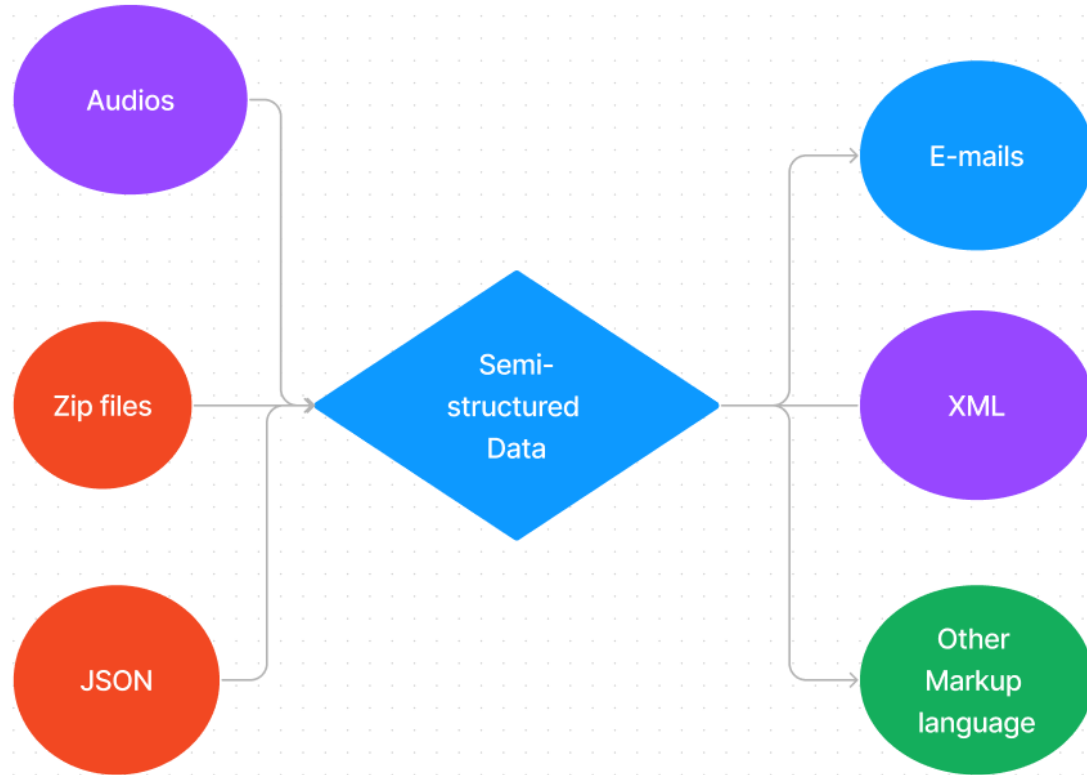
1. Structured data can only be leveraged in cases of predefined functionalities. This means that structured data has limited flexibility and is suitable for certain specific use cases only.

2. Structured data is stored in a data warehouse with rigid constraints and a definite schema. Any change in requirements would mean updating all of that structured data to meet the new needs. This is a massive drawback in terms of resource and time management.

Semi-Structured Data

- Semi-structured data is not bound by any rigid schema for data storage and handling. The data is not in the relational format and is not neatly organized into rows and columns like that in a spreadsheet. However, there are some features like key-value pairs that help in discerning the different entities from each other.
- Since semi-structured data doesn't need a structured query language, it is commonly called NoSQL data.

- A data serialization language is used to exchange semi-structured data across systems that may even have varied underlying infrastructure.
- Semi-structured content is often used to store metadata about a business process but it can also include files containing machine instructions for computer programs.
- This type of information typically comes from external sources such as social media platforms or other web-based data feeds.



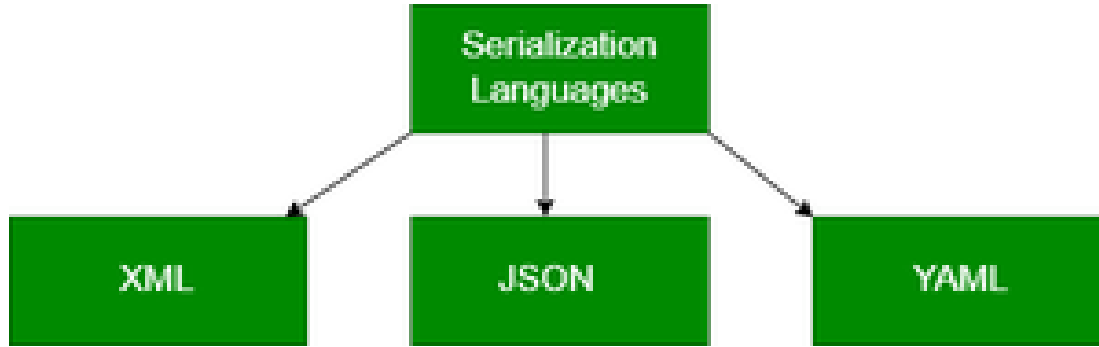
Semi-Structured Data

- Data is created in plain text so that different text-editing tools can be used to draw valuable insights. Due to a simple format, data serialization readers can be implemented on hardware with limited processing resources and bandwidth.

Data Serialization Languages

- Software developers use serialization languages to write memory-based data in files, transit, store, and parse. The sender and the receiver don't need to know about the other system. As long as the same serialization language is used, the data can be understood by

both systems comfortably. There are three predominantly used Serialization languages.



1. XML– XML stands for eXtensible Markup Language. It is a text-based markup language designed to store and transport data. XML parsers can be found in almost all popular development platforms. It

is human and machine-readable. XML has definite standards for schema, transformation, and display. It is self-descriptive. Below is an example of a programmer's details in XML.

- XML

```
<ProgrammerDetails>

  <FirstName>Jane</FirstName>

  <LastName>Doe</LastName>

  <CodingPlatforms>

    <CodingPlatform Type="Fav">GeeksforGeeks</CodingPlatform>
```

```
<CodingPlatform Type="2ndFav">Code4Eva!</CodingPlatform>

<CodingPlatform Type="3rdFav">CodeisLife</CodingPlatform>

</CodingPlatforms>

</ProgrammerDetails>

<!--The 2ndFav and 3rdFav Coding Platforms are imaginative because Geeksforgeeks
is the best!-->
```

XML expresses the data using tags (text within angular brackets) to shape the data (for ex: FirstName) and attributes (For ex: Type) to feature the data. However, being a verbose and voluminous

language, other formats have gained more popularity.

2. JSON– JSON (JavaScript Object Notation) is a lightweight open-standard file format for data interchange. JSON is easy to use and uses human/machine-readable text to store and transmit data objects.

- Javascript

```
{  
  
  "firstName": "Jane",
```

```
"lastName": "Doe",

"codingPlatforms": [

  { "type": "Fav", "value": "Geeksforgeeks" },

  { "type": "2ndFav", "value": "Code4Eva!" },

  { "type": "3rdFav", "value": "CodeisLife" }

]

}
```

This format isn't as formal as XML. It's more like a key/value pair model than a formal data depiction. Javascript has inbuilt support for JSON. Although JSON is very popular amongst web developers,

non-technical personnel find it tedious to work with JSON due to its heavy dependence on JavaScript and structural characters (braces, commas, etc.)

3. YAML– YAML is a user-friendly data serialization language. Figuratively, it stands for YAML Ain't Markup Language. It is adopted by technical and non-technical handlers all across the globe owing to its simplicity. The data structure is defined by line separation and indentation and reduces the dependency on structural characters. YAML is extremely comprehensive and its

popularity is a result of its human-machine readability.

```
firstName: Jane
lastName: Doe
CodingPlatforms:
  - type: Fav
    value: Geeksforgeeks
  - type: 2ndFav
    value: Code4Eva!
  - type: 3rdFav
    value: CodeisLife
```

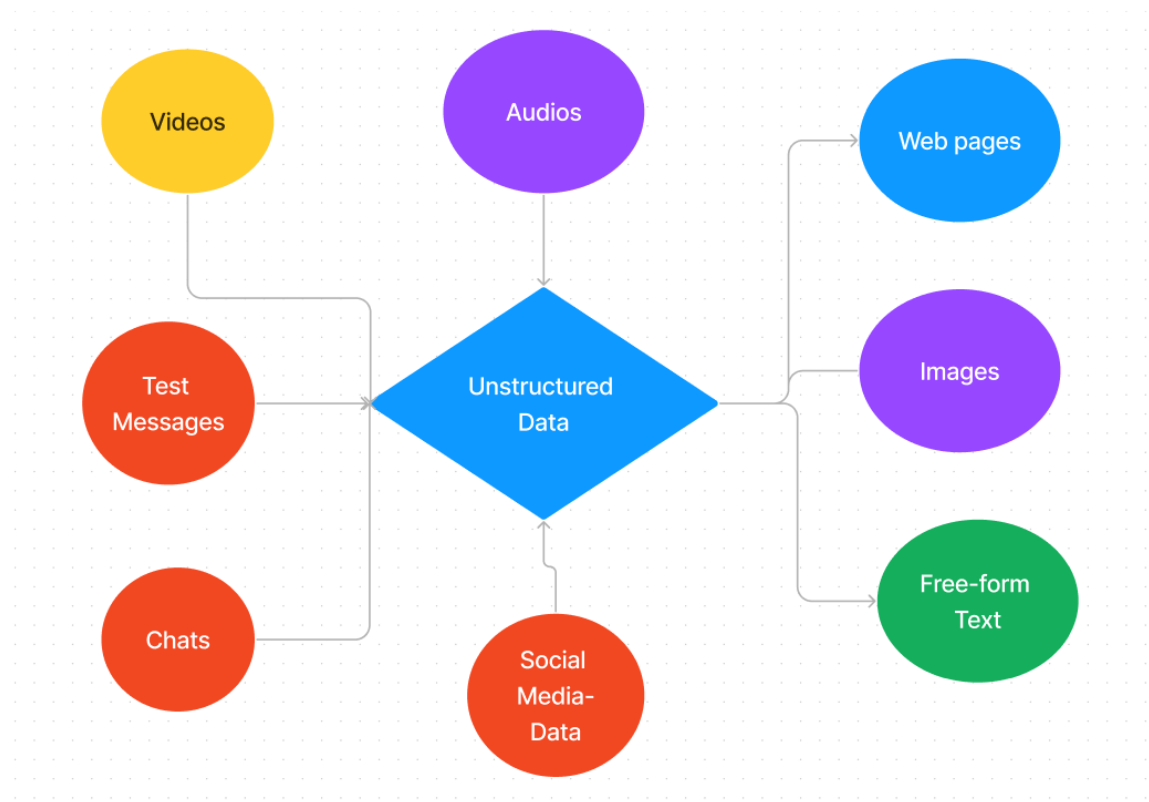
YAML example

A product catalog organized by tags is an example of semi-structured data.

Unstructured Data

- Unstructured data is the kind of data that doesn't adhere to any definite schema or set of rules. Its arrangement is unplanned and haphazard.
- Photos, videos, text documents, and log files can be generally considered unstructured data. Even though the metadata accompanying an image or a video may be semi-structured, the actual data being dealt with is unstructured.

- Additionally, Unstructured data is also known as “dark data” because it cannot be analyzed without the proper software tools.



Un-structured Data

Big Data

- Big data refers to data that is so large, fast and complex that it's difficult or impossible to process using traditional methods.
- Data that contains greater **variety**, arriving in increasing **volumes** and with more **velocity**.
- This is also known as the three Vs.

Example of Big Data

- **Stock Exchange:**

The **New York Stock Exchange** is an example of Big Data that generates about ***one terabyte*** of new trade data per day.

- **Social Media:**

The statistic shows that ***500+terabytes*** of new data get ingested into the databases of social media site **Facebook**, every day.

- **Jet engine:**

A single **Jet engine** can generate ***10+terabytes*** of data in ***30 minutes*** of flight time. With many thousand flights per day, generation of data reaches up to many ***Petabytes***.

Comparison of Big Data with Conventional Data

Big data

- Huge data sets.
- Unstructured data such as text, video, and audio.
- Hard-to-perform queries and analysis.
- Need tools such as Hadoop, Hive, Hbase, Pig and so on.
- Generated by big financial institutions, Facebook, Google, Amazon, eBay, Walmart, and so on.

Conventional data

- Data set size in control.
- Normally structured data such as numbers and categories, but it can take other forms as well.
- Relatively easy-to-perform queries and analysis.
- Tools such as SQL, SAS (Statistical Analysis System), R, and Excel alone may be sufficient.
- Generated by small enterprises and small banks.

Characteristics of Big Data

- Big data is a collection of data from many different sources and is often described by five characteristics: volume, value, variety, velocity, and veracity.



Characteristics of Big Data

Volume

- It describes the **enormous amount** of information
- Which is the defining criterion for whether we can consider a dataset as Big Data or not.

Velocity

- Velocity describes how rapidly the data is generated and how quickly it moves.
- This data flow comes from sources such as mobile phones, social media, networks, servers, etc.
- For instance, a consumer with wearable gadget that has a sensor connected to a network will keep gathering and sending data to the source.

Variety

- Variety describes the **diversity of the data** types and its heterogeneous sources.
- Big Data information draws from a vast quantity of sources, comes in three different types
 - **Structured Data:** information with a defined length and format.
 - An Excel spreadsheet with customer names, e-mails, and cities is an example of structured data.
 - **Unstructured Data:** information that can't neatly fit in the rigid, traditional row and column structure.
 - Unstructured data includes images, texts, and videos
 - **Semi-structured Data:** information that features like metadata.
 - This category includes emails, web pages, and TCP/IP packets.

Veracity

- Veracity describes the **data's accuracy and quality**.
- Gathered data could have missing pieces, may be inaccurate or may not be able to provide real, valuable insight.
- Veracity, overall, refers to the level of trust there is in the collected data.

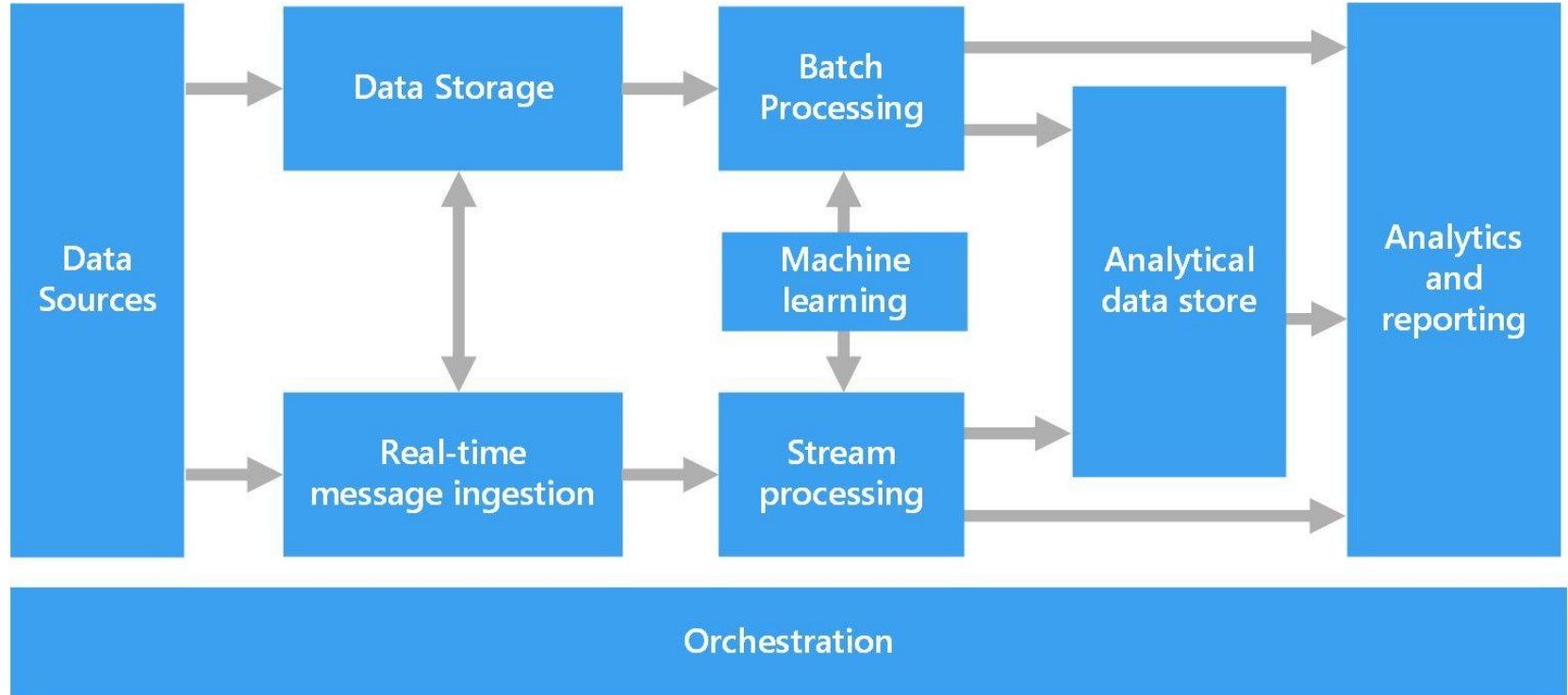
Value

- The most important characteristic from the perspective of the business.
- The value of big data usually comes from **insight discovery** and pattern recognition that lead to more effective operations, stronger customer relationships and other clear and quantifiable business benefits.
- The more insights derived from the Big Data, the higher its value.

Big Data Architecture

- Big data solutions typically involve one or more of the following types of workload:
 - Batch processing of big data sources at rest.
 - Real-time processing of big data in motion.
 - Interactive exploration of big data.
 - Predictive analytics and machine learning.
- A big data architecture is designed to handle the ingestion, processing, and analysis of data that is too large or complex for traditional database systems.

Big Data Architecture-Components



Most big data architectures include some or all of the following components:

Data sources

- All big data solutions start with one or more data sources.
- Examples include:
 - Application data stores, such as relational databases.
 - Static files produced by applications, such as web server log files.
 - Real-time data sources, such as IoT devices.

Data storage

- Data for batch processing operations is typically stored in a distributed file store that can hold high volumes of large files in various formats (data lake).
- Options for implementing this storage include Azure Data Lake Store or blob containers in Azure Storage.

Batch Processing

- It is a method computers use to periodically complete high-volume, repetitive data jobs, such as backups, filtering, and sorting.
- It can be compute intensive and inefficient to run on individual data transactions.
- Instead, data systems process such tasks in batches, often in off-peak times .
- For example
 - consider an ecommerce system that receives orders throughout the day. Instead of processing every order as it occurs, the system might collect all orders at the end of each day and share them in one batch with the order fulfillment team.

Real-time Message Ingestion

- If the solution includes real-time sources, the architecture must include a way to capture and store real-time messages for stream processing.
- This might be a simple data store, where incoming messages are dropped into a folder for processing. However, many solutions need a message ingestion store to act as a buffer for messages, and to support scale-out processing, reliable delivery, and other message queuing semantics.
- This portion of a streaming architecture is often referred to as stream buffering. Options include Azure Event Hubs, Azure IoT Hub, and Kafka.

Stream Processing

- After capturing real-time messages, the solution must process them by filtering, aggregating, and preparing the data for analysis.
- The processed stream data is then written to an output sink.
- Azure Stream Analytics provides a managed stream processing service based on perpetually running SQL queries that operate on unbounded streams.
- You can also use open source Apache streaming technologies like Spark Streaming in an HDInsight cluster.

Machine Learning

- Reading the prepared data for analysis (from batch or stream processing), machine learning algorithms can be used to build models that can predict outcomes or classify data.
- These models can be trained on large datasets, and the resulting models can be used to analyze new data and make predictions.
- This can be done using Azure Machine Learning

Analytical Data Store

- Many big data solutions prepare data for analysis and then serve the processed data in a structured format that can be queried using analytical tools.
- The analytical data store is used to save the processed data.

Analytics & Reporting

- Analytics is the technique of examining data and reports to obtain actionable insights that can be used to realise and improve business performance.
- To empower users to analyze the data, the architecture may include a data modeling layer, such as a multidimensional OLAP cube or tabular data model in Azure Analysis Services.
- It might also support self-service BI, using the modeling and visualization technologies in Microsoft Power BI or Microsoft Excel.
- Analysis and reporting can also take the form of interactive data exploration by data scientists or data analysts

Orchestration

- Most big data solutions consist of repeated data processing operations, that transform source data, move data between multiple sources and sinks, load the processed data into an analytical data store, or push the results straight to a report or dashboard.
- Orchestration technology is used to automate these workflows, tools such as Azure Data Factory or Apache Oozie and Sqoop are used for this purpose.

Advantages Of Big Data Processing

Big data processing offers a range of significant advantages, particularly as data grows in volume, velocity, and variety. Here are some key benefits:

1. Improved Decision Making

Data-Driven Insights: Big data allows organizations to make more informed, data-driven decisions by processing vast amounts of information quickly and accurately. This helps uncover trends, patterns, and correlations that might be difficult to spot with smaller datasets.

Real-Time Analysis: Big data processing allows for real-time or near-real-time analytics. This is especially useful for applications that require timely insights, such as financial markets, fraud detection, and customer sentiment analysis.

2. Scalability and Flexibility

Handling Massive Data Volumes: Traditional data processing systems often struggle with extremely large datasets. Big data frameworks (like Hadoop and Spark) are specifically designed to scale across multiple servers, enabling efficient processing of petabytes of data.

Distributed Computing: Big data processing frameworks use distributed computing, meaning that tasks can be split across multiple machines. This allows large datasets to be processed faster and more efficiently.

3. Cost Efficiency

Cost-Effective Storage: With distributed file systems like HDFS (Hadoop Distributed File System), organizations can store vast amounts of data on inexpensive hardware. Cloud storage platforms (like AWS, Google Cloud, and Azure) also offer cost-effective storage options.

Parallel Processing: Big data tools like Apache Spark enable parallel processing, where

tasks can be executed simultaneously on multiple nodes, significantly reducing processing time and improving performance.

4. Enhanced Customer Insights and Personalization

Personalized Services: Big data allows companies to collect and analyze customer data from various sources, providing insights into customer preferences, behaviors, and needs. This enables businesses to offer personalized recommendations, targeted marketing, and optimized product offerings.

Customer Segmentation: Big data processing helps businesses segment customers more effectively by analyzing a variety of factors, enabling more accurate and efficient

targeting of different market segments.

5. Predictive Analytics

Forecasting Trends: Big data can be used to build predictive models that help forecast trends and behaviors. This is particularly useful in industries like finance, retail, and healthcare, where anticipating future events or behaviors can provide a competitive edge.

Risk Management: With predictive models based on big data, businesses can identify potential risks before they occur and take preventative actions, whether it's detecting fraud, analyzing market fluctuations, or identifying machinery failures.

6. Improved Operational Efficiency

Optimizing Operations: By analyzing big data, organizations can identify inefficiencies and optimize business operations. For example, supply chain and logistics companies can optimize routes, inventory, and demand forecasts using big data analytics.

Automation and Monitoring: Big data processing allows for the automation of many tasks. Real-time monitoring and analysis can help businesses quickly identify and resolve issues, reducing downtime and improving system performance.

7. Enhanced Research and Innovation

Scientific Discovery: Big data enables researchers to process large datasets in fields like genomics, astronomy, and climate science. This accelerates discovery by allowing for more accurate models and simulations.

Innovation in Products and Services: Companies can leverage big data to identify emerging market needs and develop innovative products or services. By analyzing customer feedback, usage patterns, and market trends, organizations can drive new product development.

8. Improved Collaboration

Data Sharing Across Teams: Big data processing systems allow data to be stored, processed, and accessed in a collaborative way across different teams and departments. This helps break down silos within organizations and fosters better communication and coordination.

Integration of Various Data Sources: Big data tools enable the integration of data from diverse sources (e.g., social media, IoT devices, transactional databases), creating a more comprehensive view of the data.

9. Competitive Advantage

Faster Time to Market: Companies that process and analyze data quickly gain a competitive advantage in terms of launching products, identifying trends, and responding to market changes.

Better Business Strategy: The insights derived from big data can influence long-term business strategies. Companies that leverage big data effectively are more likely to outperform their competitors.

10. Data Security and Compliance

Better Security: Big data analytics tools can help detect suspicious activity, prevent security breaches, and analyze threat patterns in real time.

Regulatory Compliance: Big data processing helps organizations monitor compliance with regulations by analyzing large amounts of transactional and operational data and identifying potential compliance risks.

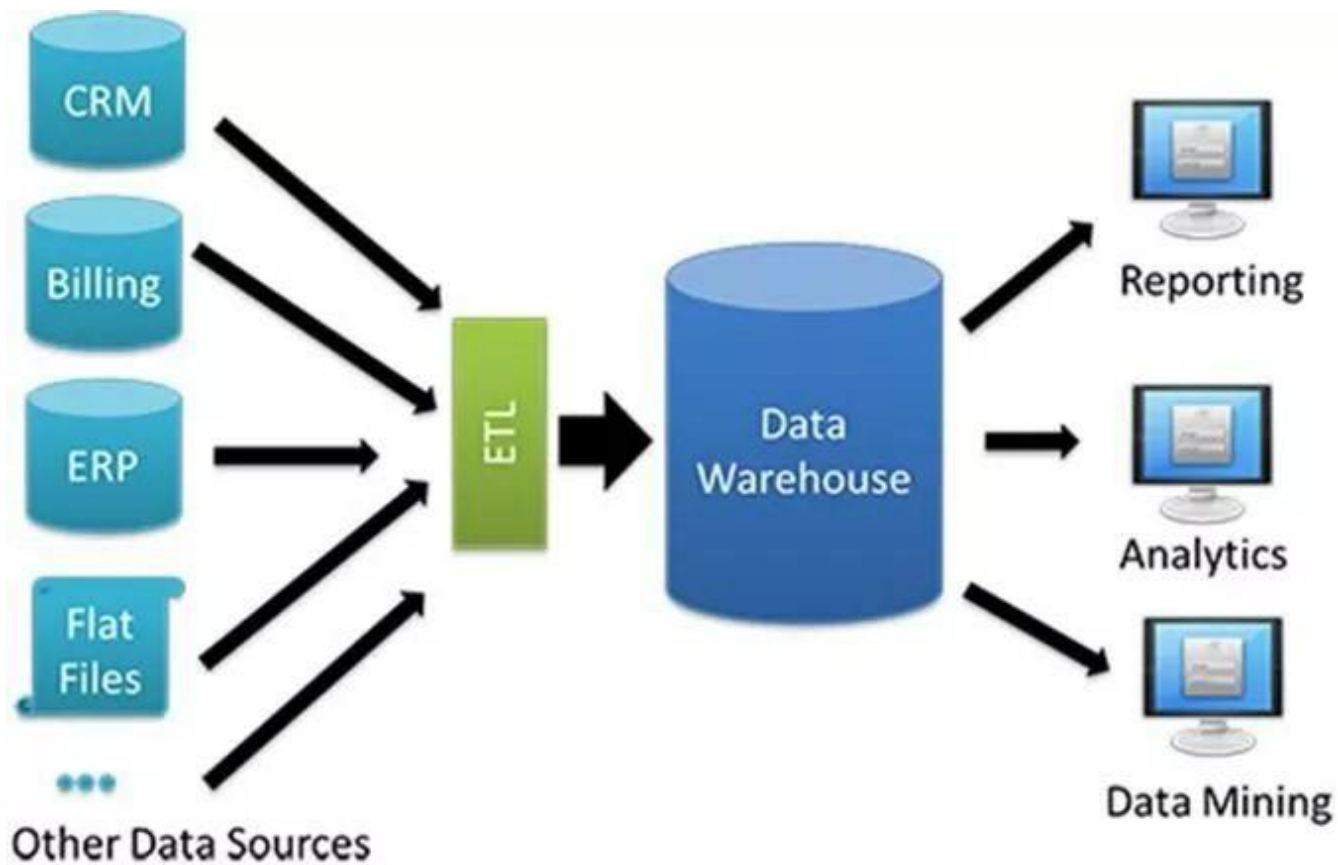
11. Advanced Machine Learning and AI

Training Machine Learning Models: Big data provides the foundation for training machine learning algorithms and building artificial intelligence (AI) models. With large datasets, these models can improve in accuracy and precision, leading to better predictions and automation.

Real-Time AI: AI and machine learning algorithms powered by big data can provide real-time insights, enabling automated decision-making in areas like customer service (e.g., chatbots) and predictive maintenance.

Data warehouse

- A data warehouse is a central repository of information that can be analyzed to make more informed decisions.
- Data flows into a data warehouse from transactional systems, relational databases, and other sources, typically on a regular interval.
- Business analysts, data engineers, data scientists, and decision makers access the data through business intelligence (BI) tools, SQL clients, and other analytics applications.



- Data and analytics have become indispensable to businesses to stay competitive.
- Business users rely on reports, dashboards, and analytics tools to extract insights from their data, monitor business performance, and support decision making.
- Data warehouses power these reports, dashboards, and analytics tools by storing data efficiently to minimize the input and output (I/O) of data and deliver query results quickly to hundreds and thousands of users concurrently.

Big Data- challenges

- Storage
- Processing
- Security
- Data Validation and Finding and Fixing Data Quality Issues.
- Scaling Big Data Systems
- Evaluating and Selecting Big Data Technologies
- Real-Time Insights

Big Data Platform

- Big Data platform is an IT solution which combines several Big Data tools and utilities into one package for managing and analyzing Big Data.
- It is an enterprise class IT platform that enables organization in developing, deploying, operating and managing a big data infrastructure /environment.
- There are several Open source and commercial Big Data Platform in the market with varied features which can be used in Big Data environment.
- Big data platform generally consists of big data storage, servers, database, big data management, business intelligence and other big data management utilities.
- The primary benefit behind a big data platform is to reduce the complexity of multiple vendors/ solutions into a one cohesive solution.

Features of Big Data Platform

- Big Data platform should be able to accommodate new platforms and tool based on the business requirement.
- It should support linear scale-out
- It should have capability for rapid deployment.
- It should support variety of data format.
- Platform should provide data analysis and reporting tools.
- It should provide real-time data analysis software.
- It should have tools for searching the data through large data sets.

List of Big Data Platforms

- b) Hadoop
- c) Cloudera
- d) Amazon Web Services
- e) Hortonworks
- f) MapR
- g) IBM Open Platform
- h) Microsoft HDInsight
- i) Intel Distribution for Apache Hadoop
- j) Databricks Enterprise Analytics
- k) Teradata Enterprise Access for Hadoop
- l) Pivotal HD

Hadoop

- "Hadoop" refers to the entire ecosystem of open-source software projects that provide a framework for distributed storage and processing of large data sets.
- It is an umbrella term that encompasses various components and tools designed to work together to solve big data challenges.
- Hadoop includes both **storage** (Hadoop Distributed File System or HDFS) and **processing** (MapReduce) frameworks, as well as additional components for data management, processing, and analysis.

Apache Hadoop

- Apache Hadoop is Big Data platform and software package which is Apache sponsored project.
- Under Apache Hadoop project various other software is being developed which runs on the top of Hadoop system to provide enterprise grade data management and analytics solutions to enterprise.
- Apache Hadoop is open-source, distributed file system which provides data processing and analysis engine for analyzing large set of data.
- Hadoop can run on Windows, Linux and OS X operating systems, but it is mostly used on Ubuntu and other Linux variants.

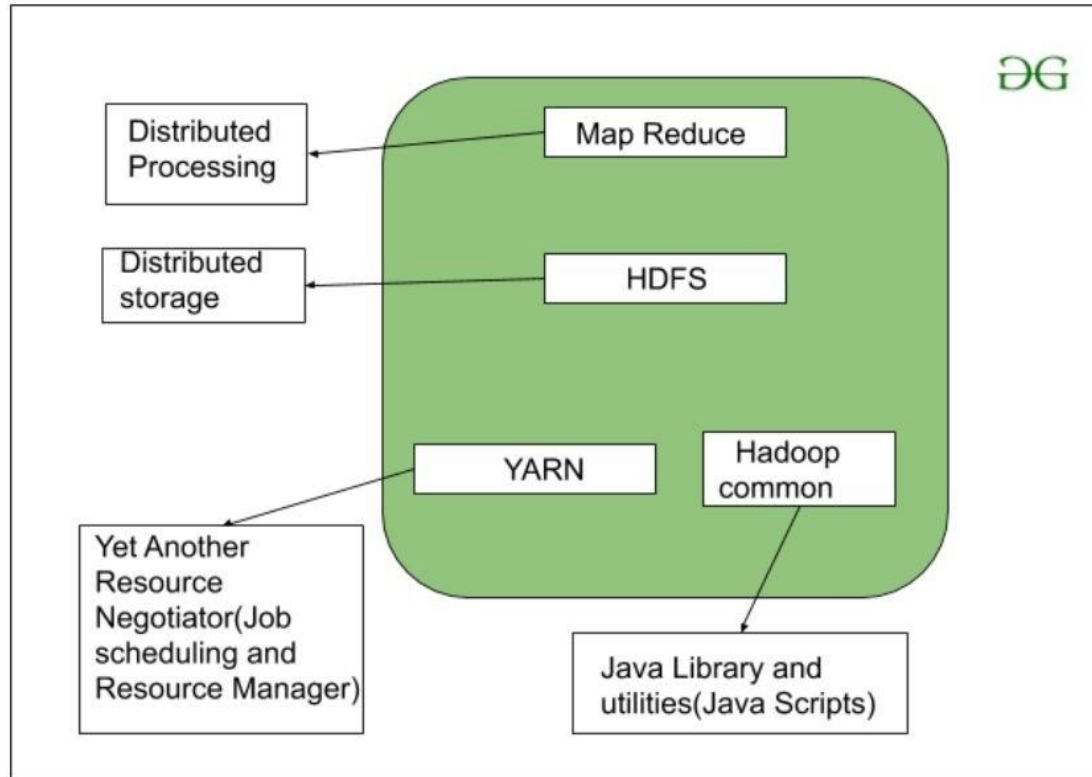
HDFS

- The Hadoop Distributed File System (HDFS) is a distributed file system designed to store and manage very large data sets reliably and fault-tolerantly across multiple commodity hardware nodes.
- It is a key component of the Apache Hadoop ecosystem and serves as the primary storage system for data in Hadoop clusters.
- Key features and characteristics of HDFS include:
 - Distributed Storage
 - Fault Tolerance
 - Master-Slave Architecture
 - Streaming Data Access
 - Designed for Commodity Hardware

MapReduce

- The MapReduce engine was originally written by Google and this is the system which enables the developers to write program which can run in parallel on 100 or even 1000s of computer nodes to process vast data sets.
- After processing all the job on the different nodes it comes the results and return it to the program which executed the MapReduce job.
- This software is platform independent and runs on the top of Hadoop ecosystem.
- It can process tremendous data at very high speed in Big Data environment.

Hadoop – Architecture



Apache Spark

- Apache Spark is software that runs on the top of Hadoop and provides API for real-time,in-memory processing and analysis of large set of stored in the HDFS.
- It stores the data into memory for faster processing.
- Apache Spark runs program 100 times faster in-memory and 10 times faster on disk as compared to the MapReduce.
- Apache Spark is here to faster the processing and analysis of big data sets in Big Data environment.
- Apache Spark is being adopted very fast by the business to analyze their data set to get real value of their data.

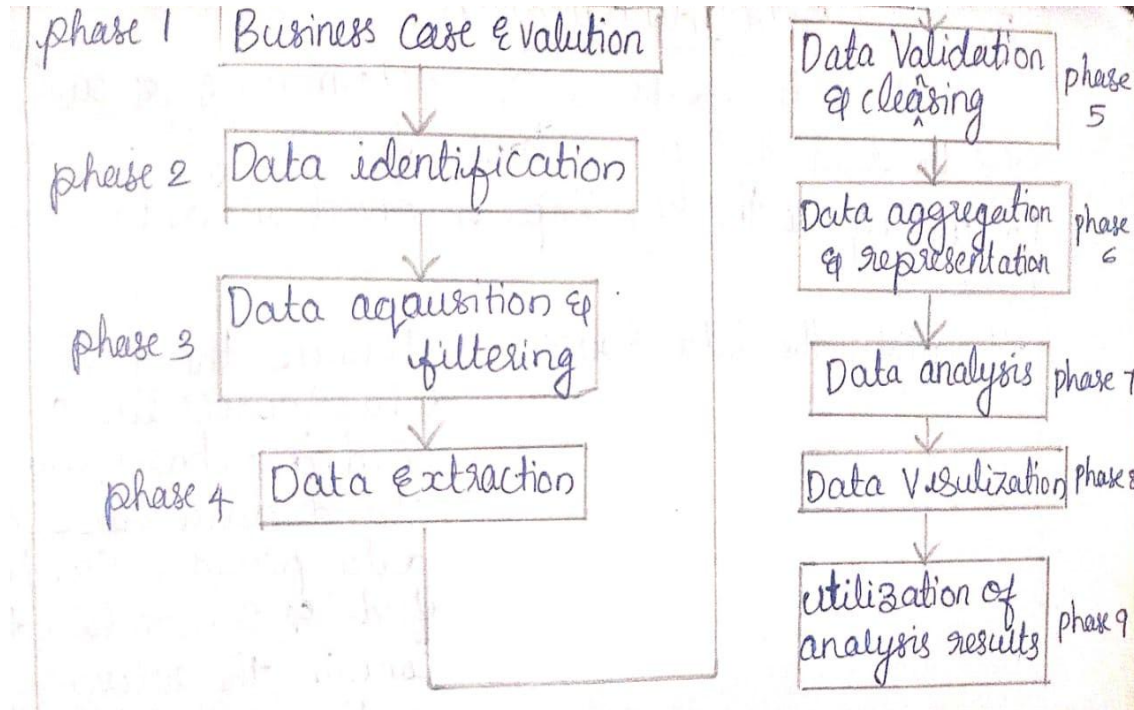
Big data Analytics Lifecycle

Bigdata Analytics lifecycle refers to the process of extracting valuable insights and knowledge from large and complex data sets. It involves several stages, from data collection and storage to analysis and decision making.

The Big Data Analytics Life cycle is divided into nine phases, named as :

1. Business Case/Problem Definition (Business case evaluation)
2. Data Identification
3. Data Acquisition and filtration
4. Data Extraction

5. Data Validation and Cleaning
6. Data Aggregation & Representation(Storage)
7. Exploratory Data Analysis
8. Data Visualization(Preparation for Modeling and Assessment)
9. Utilization of analysis results.



Phase 1: Business case evaluation

Business case evaluation is the first phase of the Bigdata Analytics Lifecycle. It involves assessing the potential value and feasibility of a data analytics project before committing resources to it. Following are the key considerations and steps involved in evaluating a business case in the context of Bigdata

- Define the problem: Clearly define the business problem that the analytics project aims to address
- Set Objectives: Establish quantifiable objectives that can be used to evaluate the success of the project.
- Identify stakeholders: Determine business leaders, domain experts, data scientists, IT professionals, and end users who will involve in the evaluation process.

- Assess data availability: Evaluate the availability and quality of the data required to address the business problem

Phase 2: Data Identification

It is the process of determining and understanding the data that is available for analysis. Following are the key steps involved in data identification within the data analytics lifecycle.

- Identify the data sources: determine the potential data sources like structured databases, unstructured data files, external data providers, social media feeds and sensor data that contain the relevant information needed for analysis.
- Assess data quality: Evaluate the quality of the data sources to ensure they meet the requirements

- Determine data relevance: Assess the data's relevance in terms of the variables, attributes or features it contains that are required to address the analytics questions.
- Evaluate data availability: determine if the identified data sources are available and accessible for analysis. Consider any legal or contractual constraints that may affect the use of certain data sources.
- Perform data profiling: data profiling helps in understanding that data's characteristics and assisting in subsequent data pre-processing steps.

Phase 3: Data Acquisition and Filtering

It involves collecting and preparing the relevant data for analysis.

Key points involved in data acquisition are:

- Data sources: identify potential data sources to analyse data
- Data collection: Extract the data from the identified sources. This can involve techniques like web scrapping, data feeds, data streaming or direct database queries
- Data integration: Combine the data from multiple sources into a unified data set. This may involve transforming the data into a common format or resolving inconsistencies.

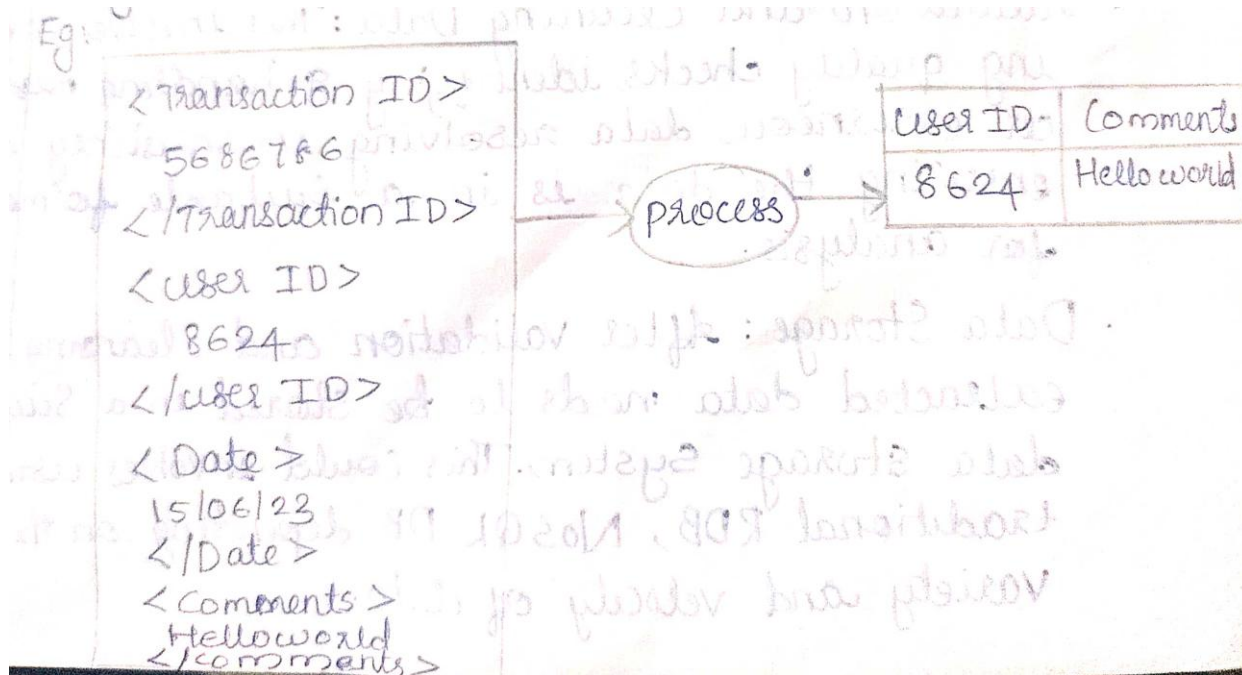
Key points involved in data filtering are:

- Data quality assessment: evaluate the quality of acquired data. this involves examining factors like completeness, accuracy, consistency, and relevance.
- Data cleaning: perform data cleaning operations to handle missing values, outliers, duplicates and inconsistencies.
- Data reduction: reduce the dataset size if needed, especially when dealing with large volumes of data.
- Data transformation: convert the data into a suitable format for analysis.

Phase 4: Data Extraction

The data extraction life cycle phase is dedicated to extracting disparate data and transforming it into a format that the underlying bigdata solution can use for the

purpose of the data analysis.



- Identify data sources: identify the relevant data sources that contain the required information for analysis.
- Define Requirements: this involves determining the scope of the data extraction, selecting the data attributes to be extracted specifying the time period of interest and establishing any filtering or transformation requirements.
- Choose extraction techniques: depending on the nature of data sources, different extraction techniques may be employed. For structured database, sql queries can be used to extract data based on specific criteria. For unstructured or semi structured data, web scrapping, data passing or natural language processing (NLP) may be employed to extract relevant information.

Phase 5: Data Validation and Cleaning

Data validation and cleaning are critical steps in the big data analytics lifecycle. They help ensure the accuracy, consistency and reliability of the data used for analysis. Key points involved in data validation and cleaning are:

- Data acquisition: Data validation can involve checking the integrity and completeness of the acquired data to ensure it meets the required quality standards.
- Data pre-processing: this involves data cleaning, integration, transformation and reduction
- Data storage: cleaned and pre-processed data is then stored in a suitable data storage system.

- Data analysis: once the data is stored, it can be analysed using various techniques such as statistical analysis, data mining, ML or visualization.

Phase 6: Data Aggregation and Representation

This phase is dedicated to integrating multiple datasets to arrive at a unified view.

Key points involved are:

- Data aggregation: By effectively aggregating and representing data, organizations can derive valuable insights and make informed decisions. In the bigdata analytics life cycle, data aggregation typically occurs using data ingestion, data integration, data cleansing, data transformation and data storage.

- Data representation: once the data is aggregated, it needs to be represented in a suitable format using data modelling, dimensional modelling, data visualization, summary statistics, and machine readable formats.

Phase 7: Data Analysis

It is the iterative process that involves extracting meaningful insights from large and complex datasets. The study is repeated until a proper pattern or association is discovered. Key points involve in data analysis are:

- Exploratory data analysis: in this phase, analyst performs preliminary data exploration to gain a deeper understanding of the data set. They do descriptive statistics data visualization using charts and graphs, pattern recognition, correlation and outliers.

- Data transformation and feature engineering: this involves creating new features, aggregating data, normalizing and convert the categorical variables into numerical representation.
- Insight and visualization: after obtaining the final mode, analyst interpret the results and extract actionable insights from the data.
- Integration: once the analysis is completed, the insights and models are integrated into the existing systems to enable decision-making or automat certain tasks
- Monitoring and Maintenance: data analysis is an ongoing process and it is essential to continuously monitor the performance of deployed models and maintain their accuracy.

Phase 8: Data Visualization

This phase is dedicated to do the data analysis using data visualization techniques and tools to communicate the analysis results effectively to business users. The outcomes of data visualization provide the ability to perform visual analysis to its users. It helps analyst, data scientist, and business users make sense of the data, identify relationships and make data driven decisions. Key points involved in data visualization are:

- Data exploration and preparation: once the data is acquired, it needs to be explored and prepared for analysis. Data visualization techniques like scatter plots, histograms and other maps can be used to gain an initial understanding of the data distribution, identify outliers and detect patterns.

- Data modelling and analysis: data visualization helps in understanding the output of these models by visually representing the results
- Data visualization and communication: data visualization techniques such as charts, graphs and interactive visualization help convey complex information in a clear and concise manner.
- Decision making and action: data visualization empower decision makers to understand complex patterns, trends, and dependencies facilitating effective decision making.

Phase 9: Utilization of Analysis Results

The analysis is done; the results are visualized; now it's time for the business users to make decisions to utilize the results. The results can be used for

optimization, to refine the business process. It can also be used as an input for the systems to enhance performance.

Data Analytics Processes

Data Collection: Data is the heart of Big Data Analytics. It is the process of the collection of data from various sources, which can include customer reviews, surveys, sensors, social media etc. The main goal of data collection is to gather as much relevant data as possible. The more data, the richer the insights.

Data Cleaning (Data Preprocessing): Once we have the data, it often needs some cleaning. This process involves identifying and dealing with missing values, correcting errors, and removing duplicates. It's like sifting through a treasure trove to remove any rocks or debris,

leaving only the valuable gems behind.

Data Processing: Next, we need to process the data. This involves different steps like organizing, structuring, and formatting it in a way that makes it appropriate for analysis. Think of it like a chef preparing ingredients before cooking. Data processing makes the raw data more digestible for analytics tools.

Data Analysis: Data analysis is performed using various statistical, mathematical, and machine learning techniques to extract valuable insights from the processed data. For instance, it can reveal customer preferences, market trends, or patterns in healthcare data.

Data Visualization: Data analysis results are often presented in the form of visualizations – charts, graphs, and interactive dashboards. These visual representations make complex data easy to understand and enable decision-makers to see trends and patterns at a glance.

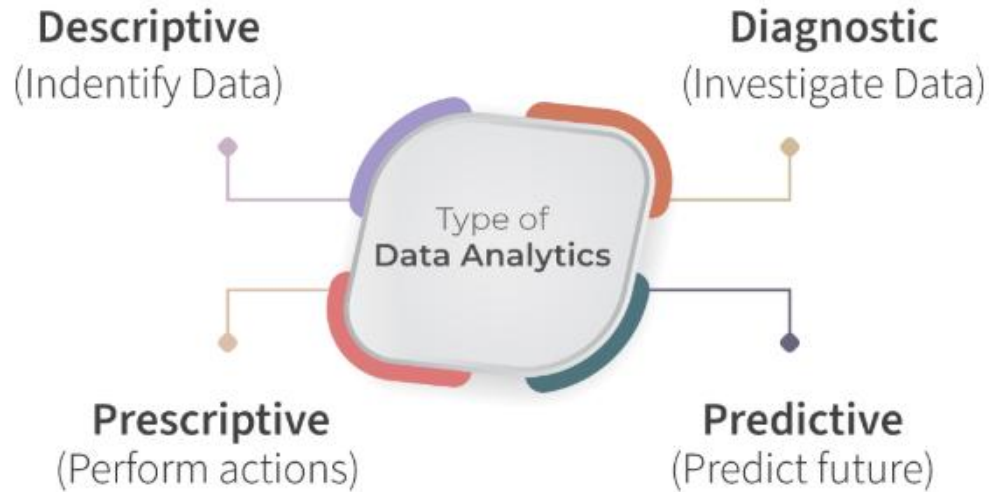
Data Storage and Management: Storing and managing the analyzed data is crucial. It's like archiving your findings. You may want to revisit the insights in the future, and well-organized storage is essential for that. Additionally, it's important to ensure data security and compliance with regulations during this critical step.

Continuous Learning and Improvement: Big Data Analytics isn't a one-time affair. It's an ongoing process. As you collect and analyze more data, you learn more about your operations or customers. This insight can lead to refining your data collection methods and analysis techniques for better results. Big Data Analytics is about collecting, cleaning, processing, and analyzing data to uncover valuable insights. It's a multi-step process that transforms raw data into fruitful insights. By following these steps and continually improving your analytics approach, you can use the power of data to make informed decisions and gain a competitive edge.

Data Analytics Methods and Techniques

Data analysis techniques have significantly evolved, providing a comprehensive toolkit for understanding, interpreting, and predicting data patterns. These methods are crucial in extracting actionable insights from data, enabling organizations to make informed decisions

Type of Data Analytics



Descriptive Data Analysis

Descriptive analysis is considered the beginning point for the analytic journey and often strives to answer questions related to what happened. This technique follows ordering factors, manipulating and interpreting varied data from diverse sources, and then turning it into valuable insights.

In addition, conducting this analysis is imperative as it allows individuals to showcase insights in a streamlined method. This technique does not allow you to estimate future outcomes - such as identifying specific reasoning for a particular factor. It will keep our data streamlined and simplify to conduct a thorough evaluation for further circumstances.

Examples of Descriptive Data Analysis :

Sales Performance: A retail company might use descriptive statistics to understand the average sales volume per store or to find which products are the best sellers.

Customer Satisfaction Surveys: Analyzing survey data to find the most common responses or average scores.

Qualitative Data Analysis

Qualitative data analysis techniques cannot be measured directly, and hence, this technique is utilized when an organization needs to make decisions based on subjective interpretation. For instance, qualitative data can involve evaluating

customer feedback, the impact of survey questions, the effectiveness of social media posts, analyzing specific changes or features of a product, and more.

The focus of this technique should be identifying meaningful insights or answers from unstructured data such as transcripts, vocal feedback, and more. Additionally, qualitative analysis aids in organizing data into themes or categories, which can be further automated. Quantitative data analysis refers to measurable information, which includes specific numbers and quantities. For instance, sales figures, email campaigns based on click-through rates, website visitors, employee performance percentage, or percentage for revenue generated, and more.

Examples of Qualitative Data Analysis:

Market Analysis: A business might analyze why a product's sales spiked in a particular quarter by looking at marketing activities, price changes, and market trends.

Medical Diagnosis: Clinicians use diagnostic analysis to understand the cause of symptoms based on lab results and patient data.

Predictive Data Analysis

Predictive data analysis allows us to forecast future events by addressing enquiries regarding potential outcomes. Individuals need to utilize the results of descriptive data analysis, exploratory and diagnostic analysis techniques, and combine machine

learning and artificial intelligence. Using this method, we can get an overview of future trends and identify potential issues and loopholes in your dataset.

In addition, we can discover and develop initiatives to enhance varied operation processes and your competitive edge with insightful data. With easy-to-understand insights, businesses can tap into trends, common patterns, or reasons for a specific event, making initiatives or decisions for further strategies easier.

Examples of Predictive Data Analysis:

Credit Scoring: Financial institutions use predictive models to assess a customer's likelihood of defaulting on a loan.

Weather Forecasting: Meteorologists use predictive models to forecast weather conditions based on historical weather data.

Diagnostic Data Analysis

Understanding the reasons behind an occurrence simplifies the process of determining the "How" related to that particular aspect. For instance, with diagnostic analysis, we can identify why our sales results are declining and eventually explore the exact factors that led to this loss.

In addition, this technique offers actionable answers to our specific questions. It is also the most commonly preferred method in research for varied domains.

Example of Diagnostic Data Analysis:

Inventory Analysis: Checking if lower sales correlate with stock outs or overstock situations.

Promotion Effectiveness: Analyzing the impact of different promotional campaigns to see which failed to attract customers.

Regression Analysis

This method utilizes historical data to understand the impact on the dependent variable's value when one or more independent variables tend to change or remain the same. In addition, determining each variable's relationship and past development or initiative enables you to predict potential outcomes in the future. And the technique gives you the right path to make informed decisions effectively.

Let's assume we conducted a Regression Analysis for our sales report in 2022, and the results represented variables like customer services, sales channels, marketing campaigns, and more that affected the overall results. Then, we can conduct another regression analysis to check if the variables changed over time or if new variables are

impacting your sales result in 2023. By following this method, our sales can increase with improved product quality or services

Example of Regression Analysis:

Market Trend Assessment: Evaluating how changes in the economic environment (e.g., interest rates) affect property prices.

Predictive Pricing: Using historical data to predict future price trends based on current market dynamics.

Cohort Analysis

Cohort analysis includes historical data to analyze and compare specific segments in user behavior and groups a few aspects with other similar elements. This technique can provide an idea of your customer's and target audience's evolving needs.

In addition, we can utilize Cohort analysis to determine a marketing campaign's impact on certain audience groups. For instance, we can implement the features of the Cohort analysis technique to evaluate two types of email campaigns—commonly termed A/B Testing over time—and understand which variation turned out to be responsive and impactful in terms of performance.

Example of Cohort Analysis:

Customer Retention: Measuring how long newly acquired customers continue to make purchases compared to those not enrolled in the loyalty program.

Program Impact: Determining if and how the loyalty program influences buying patterns and average spend per purchase.

Factor Analysis

Factor data analysis defines the variations with observed related variables based on lower unobserved variables termed factors. In short, it helps in extracting independent variables, which is considered ideal for optimizing specific segments.

For instance, if we have a product and collect customer feedback for varied purposes, this analysis technique aids in focusing on specific factors like current trends, layout, product performance, potential errors, and more. The factors can vary depending on what we want to monitor and focus on. Lastly, factor analysis simplifies summarizing related factors in similar groups.

Example of Factor Analysis :

Service Improvement: Identifying key factors such as wait time, staff behavior, and treatment outcome that impact patient satisfaction.

Resource Allocation: Using these insights to improve areas that significantly affect patient satisfaction.

Time Series Analysis

A time series analysis technique checks data points over a certain timeframe. we can utilize this method to monitor data within a certain time frame on a loop; however, this technique isn't ideal for collecting data only in a specific time interval.

This technique is ideal for determining whether the variable changed amid the evaluation interval, how each variable is dependent, and how the result was achieved for a specific aspect. Additionally, we can rely on time series analysis to determine market trends and patterns over time. we can also use this method to forecast future events based on certain data insights.

Example of Time Series Analysis :

Demand Forecasting: Estimating sales volume for the next season based on historical sales data during similar periods.

Resource Planning: Adjusting production schedules and inventory levels to meet anticipated demand.

Cluster Analysis

Cluster analysis describes data and identifies common patterns. It is often used when data needs more evident labels or insights or has ambiguous categories. This process includes recognizing similar observations and grouping those aspects to

create clusters, which means assigning names and categorizing groups.

In addition, this technique aids in identifying similarities and disparities in databases and presenting them in a visually organized method to seamlessly compare factors. Box plot visualization is mainly preferred to showcase data clusters.

Example of Cluster Analysis:

Market Segmentation: Dividing customers into groups that exhibit similar behaviors and preferences for more targeted marketing.

Campaign Customization: Designing unique marketing strategies for each cluster to maximize engagement and conversions.

Big Data analytics tools

Here's a list of the main Big Data analytics tools used for processing, analyzing, and gaining insights from large datasets. These tools are designed to handle massive volumes of structured and unstructured data:

1. Apache Hadoop

- Category: Data storage and processing framework
- Use: Distributed storage and processing of large datasets.
- Description: An open-source framework for storing and processing large datasets using a distributed approach. It consists of the Hadoop Distributed File System (HDFS) for storage and MapReduce for processing.

2. **Apache Spark**

- Category: Data processing engine
- Use: Real-time data processing and analytics.
- Description: A fast, in-memory processing engine that supports both batch and real-time processing. It's much faster than Hadoop's MapReduce and is often used for machine learning, data streaming, and analytics at scale.

3. **Apache Flink**

- Category: Stream processing engine
- Use: Real-time data streaming and batch processing.

- Description: An open-source stream processing framework capable of handling real-time data analytics. Flink provides low-latency, high-throughput data streaming and is often used for big data analytics in real-time.

4. **Apache Kafka**

- Category: Distributed streaming platform
- Use: Real-time data ingestion and processing.
- Description: A distributed messaging system that allows you to publish, subscribe, store, and process streams of records in real-time. It is highly scalable and fault-tolerant, making it ideal for handling massive amounts of data in real-time.

5. Apache Hive

- Category: Data warehouse system
- Use: Data summarization, querying, and analysis on Hadoop.
- Description: A data warehouse built on top of Hadoop, providing a SQL-like query language (HiveQL) for querying large datasets stored in Hadoop. It's useful for big data processing and analyzing structured data with ease.

6. Apache HBase

- Category: NoSQL database
- Use: Real-time random access to big data.

- Description: A distributed, column-oriented NoSQL database that runs on top of Hadoop. HBase is ideal for handling large amounts of sparse data and is used for real-time read/write access to data.

7. **Apache Cassandra**

- Category: NoSQL database
- Use: High availability, scalability, and fault tolerance.
- Description: A highly scalable NoSQL database used for managing large datasets across many commodity servers. It's designed to handle large amounts of data across multiple nodes with no single point of failure.

8. **Google BigQuery**

- Category: Data warehouse and analytics tool
- Use: Fast SQL querying on large datasets.
- Description: A fully-managed, serverless data warehouse from Google Cloud that allows users to run SQL queries on massive datasets quickly and efficiently. It is optimized for analytical workloads.

9. **Amazon Redshift**

- Category: Data warehouse
- Use: Data warehousing and large-scale analytics.

- Description: A cloud-based data warehousing service from AWS, designed to handle large volumes of data. Redshift is optimized for running complex queries and performing analytics on structured and semi-structured data.

10. **Talend**

- Category: Data integration
- Use: Data extraction, transformation, and loading (ETL).
- Description: An open-source data integration tool for performing ETL operations on large datasets. Talend helps in extracting data from multiple sources, transforming it into a usable format, and loading it into data warehouses for analysis.

11. Databricks

- Category: Unified analytics platform
- Use: Data engineering, machine learning, and analytics.
- Description: Built around Apache Spark, Databricks offers a collaborative environment for data engineering and machine learning. It's used to process large datasets, build data pipelines, and develop machine learning models.

12. Presto

- Category: Distributed SQL query engine
- Use: Fast SQL analytics on large datasets.

- Description: An open-source distributed SQL query engine designed for running interactive queries on large datasets. Presto is optimized for low-latency queries and can query data from multiple sources, including HDFS, Amazon S3, and relational databases.

13. QlikView/Qlik Sense

- Category: Business Intelligence (BI) tools
- Use: Data visualization and analysis.
- Description: Qlik's analytics tools allow users to explore large datasets interactively, uncover insights, and create visualizations. They are popular for self-service BI, allowing users to quickly analyze and visualize data.

14. **Looker**

- Category: Data exploration and business intelligence (BI)
- Use: Data exploration, reporting, and visualization.
- Description: A modern BI tool used to analyze large datasets, build custom reports, and share insights. Looker integrates with big data platforms like Google BigQuery and Amazon Redshift, providing interactive dashboards and data exploration.

15. **Cloudera**

- Category: Data management and analytics platform
- Use: Data warehousing, machine learning, and big data analytics.

- Description: Cloudera provides an integrated suite of big data tools, including Apache Hadoop, Apache Spark, and others, to help businesses process, analyze, and manage large datasets. It also offers machine learning capabilities for data scientists and engineers.

Intelligent Data Analysis

In order to address the difficulties encountered by traditional systems in retrieving valuable information, we employ intelligent data analysis. IDA, or Intelligent Data Analysis, is a data analysis technique that relies on artificial intelligence, machine learning, high performance computers, pattern recognition, statistics, and database

and visualisation technology. IDA is a tool that use advanced techniques to analyse vast quantities of data in order to get valuable insights and information.

Data: Data is a comprehensive collection of information that encompasses a wide range of subjects and details. For instance, various sources of data include CCTV cameras and satellite photos, cell phone data, battery status, date and time, and calendar records. Social media sites such as instagram and facebook contain a substantial amount of data, including posts, descriptions, likes, dislikes, comments, and timestamps. Vast amounts of data are readily accessible and may be effortlessly gathered and acquired. However, it is not yielding any concise information.

Knowledge: Knowledge is an element of data that provides detailed information on patterns, structures, laws, principles, and other related concepts. Acquiring knowledge enables us to make accurate predictions and forecasts. Information is significantly more useful than raw data.

When dealing with enormous quantities of data, humans are unable to identify the information or extract the relevant knowledge that the data is attempting to convey. Therefore, in that scenario, we require intelligent data analysis. IDA employs artificial intelligence and machine learning techniques to analyse and convert large datasets into intelligent data insights. Decision-makers can then utilise this information to improve their services and investments. It helps in understanding the data, examining previous data, and extracting insights to recognise potential patterns.

IDA consists of three distinct stages.

1. Data preparation
2. Data mining
3. Result validation & explanation

Data preparation: Data preparation involves selecting the required data from the relevant data source and integrating it into a data set. Data should also include any past records, results, or statistics. We can utilize this information for data mining purposes.

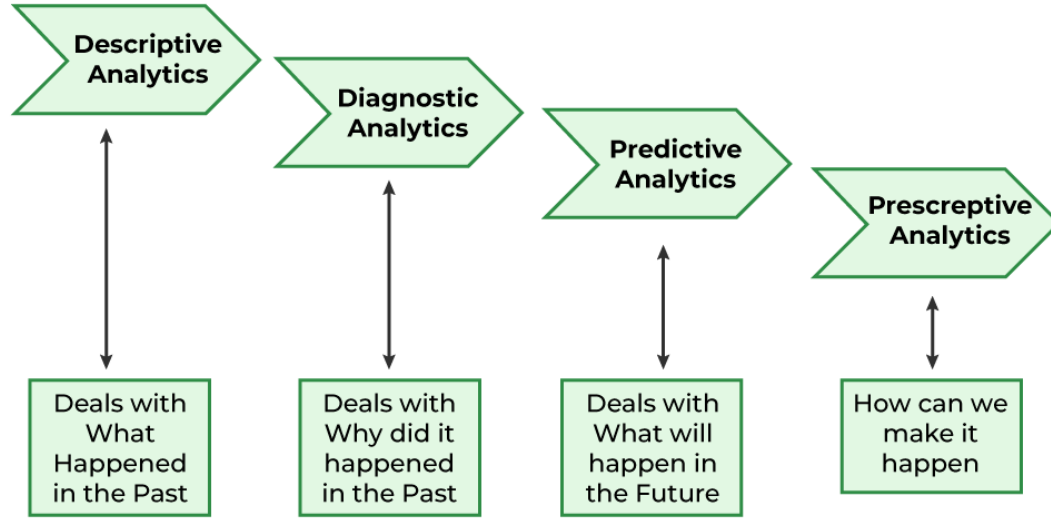
Data mining: Data mining involves the process of preparing data for usage or further study. Data mining utilises extensive databases to produce novel knowledge.

Result validation and Explanation: result validation involves the verification of patterns produced from data mining algorithms. The result explanation involves the intuitive communication of the result.

Types of Intelligent Data Analysis

There are four major types of data analytics:

1. Predictive (forecasting)
2. Descriptive (business intelligence and data mining)
3. Prescriptive (optimization and simulation)
4. Diagnostic analytics



Data Analytics and its Types

Predictive Analytics

Predictive analytics turn the data into valuable, actionable information. predictive analytics uses data to determine the probable outcome of an event or a likelihood of a situation occurring. Predictive analytics holds a variety of statistical techniques from modeling, machine learning, data mining, and game theory that analyze current and historical facts to make predictions about a future event. Techniques that are used for predictive analytics are:

- Linear Regression
- Time Series Analysis and Forecasting

- Data Mining

Basic Cornerstones of Predictive Analytics

- Predictive modeling
- Decision Analysis and optimization
- Transaction profiling

Descriptive Analytics

Descriptive analytics looks at data and analyze past event for insight as to how to approach future events. It looks at past performance and understands the performance by mining historical data to understand the cause of success or failure in

the past. Almost all management reporting such as sales, marketing, operations, and finance uses this type of analysis.

The descriptive model quantifies relationships in data in a way that is often used to classify customers or prospects into groups. Unlike a predictive model that focuses on predicting the behavior of a single customer, Descriptive analytics identifies many different relationships between customer and product.

Common examples of Descriptive analytics are company reports that provide historic reviews like:

- Data Queries
- Reports

- Descriptive Statistics
- Data dashboard

Prescriptive Analytics

Prescriptive Analytics automatically synthesize big data, mathematical science, business rule, and machine learning to make a prediction and then suggests a decision option to take advantage of the prediction.

Prescriptive analytics goes beyond predicting future outcomes by also suggesting action benefits from the predictions and showing the decision maker the implication of each decision option. Prescriptive Analytics not only anticipates what will happen and

when to happen but also why it will happen. Further, Prescriptive Analytics can suggest decision options on how to take advantage of a future opportunity or mitigate a future risk and illustrate the implication of each decision option.

For example, Prescriptive Analytics can benefit healthcare strategic planning by using analytics to leverage operational and usage data combined with data of external factors such as economic data, population demography, etc.

Diagnostic Analytics

In this analysis, we generally use historical data over other data to answer any question or for the solution of any problem. We try to find any dependency and

pattern in the historical data of the particular problem.

For example, companies go for this analysis because it gives a great insight into a problem, and they also keep detailed information about their disposal otherwise data collection may turn out individual for every problem and it will be very time-consuming. Common techniques used for Diagnostic Analytics are:

- Data discovery
- Data mining
- Correlations